

Yaru Niu

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Education

Carnegie Mellon University

Ph.D. in Mechanical Engineering, Safe AI Lab

Advisor: Prof. Ding Zhao

Pittsburgh, PA

Aug. 2022 – Present

Georgia Institute of Technology

M.S. in Electrical and Computer Engineering, CORE Robotics Lab

Advisor: Prof. Matthew Gombolay

Atlanta, GA

Aug. 2019 – April 2022

South China University of Technology (SCUT)

B.Eng. in Intelligence Science and Technology

Advisor: Prof. Zhijun Zhang

Guangzhou, China

Sep. 2015 – June 2019

University of California, Irvine

Visiting Student in the Department of EECS

Irvine, CA

June 2018 – Aug. 2018

University of California, Berkeley

Exchange Student, Concentration in Computer Science

Berkeley, CA

Aug. 2018 – Dec. 2018

Publications

(* indicates equal contributions, [Google Scholar](https://scholar.google.com/))

- Hao Zhang, **Yaru Niu**, Yikai Wang, Ding Zhao, H. Eric Tseng. HALyPO: Heterogeneous-Agent Lyapunov Policy Optimization for Human-Robot Collaboration. *International Conference on Machine Learning (ICML)*, 2026 (**Spotlight**).
- Dongsu Lee, Daehee Lee, **Yaru Niu**, Honguk Woo, Amy Zhang, Ding Zhao. Team Coordination by Interactive World Latent. *ARLET Workshop at Neural Information Processing Systems (NeurIPS)*, 2025 (**Oral**).
- Yaru Niu***, Yunzhe Zhang*, Mingyang Yu, Changyi Lin, Chenhao Li, Yikai Wang, Yuxiang Yang, Wenhao Yu, Tingnan Zhang, Zhenzhen Li, Jonathan Francis, Bingqing Chen, Jie Tan, Ding Zhao. Human2LocoMan: Learning Versatile Quadrupedal Manipulation with Human Pre-Training. *Robotics: Science and Systems (RSS)*, 2025.
- Yuyou Zhang, Yihang Yao, Shiqi Liu, **Yaru Niu**, Changyi Lin, Yuxiang Yang, Wenhao Yu, Tingnan Zhang, Jie Tan, Ding Zhao. QuietPaw: Learning Quadrupedal Locomotion with Versatile Noise Preference Alignment. *International Conference on Intelligent Robots and Systems (IROS)*, 2025.
- Yuming Feng*, Chuye Hong*, **Yaru Niu***, Shiqi Liu, Yuxiang Yang, Wenhao Yu, Tingnan Zhang, Jie Tan, Ding Zhao. Learning Multi-Agent Loco-Manipulation for Long-Horizon Quadrupedal Pushing. *International Conference on Robotics and Automation (ICRA)*, 2025.
- Changyi Lin, Xingyu Liu, Yuxiang Yang, **Yaru Niu**, Wenhao Yu, Tingnan Zhang, Jie Tan, Byron Boots, Ding Zhao. LocoMan: Advancing Versatile Quadrupedal Dexterity with Lightweight Loco-Manipulators. *International Conference on Intelligent Robots and Systems (IROS)*, 2024.
- Yuyou Zhang, **Yaru Niu**, Xingyu Liu, Ding Zhao. COMPOSER: Scalable and Robust Modular Policies for Snake Robots. *International Conference on Robotics and Automation (ICRA)*, 2024.
- Haohong Lin, Wenhao Ding, Zuxin Liu, **Yaru Niu**, Jiacheng Zhu, Yuming Niu, Ding Zhao. Safety-aware Causal Representation for Trustworthy Reinforcement Learning in Autonomous Driving. *Robotics and Automation Letters (RA-L)*, 2024.
- Yaru Niu**, Shiyu Jin*, Zeqing Zhang*, Jiacheng Zhu, Ding Zhao, Liangjun Zhang. GOATS: Goal Sampling Adaptation for Scooping with Curriculum Reinforcement Learning. *International Conference on Intelligent Robots and Systems (IROS)*, 2023.
- Mengdi Xu, Peide Huang, **Yaru Niu**, Visak Kumar, Jieli Qiu, Chao Fang, Kuan-Hui Lee, Xuewei Qi, Henry Lam, Bo Li, Ding Zhao. Group Distributionally Robust Reinforcement Learning with Hierarchical Latent Variables. *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2023.

11. Lingfeng Sun*, Chen Tang*, **Yaru Niu**, Enna Sachdeva, Chiho Choi, Teruhisa Misu, Masayoshi Tomizuka, Wei Zhan. Domain Knowledge Driven Pseudo Labels for Interpretable Goal-conditioned Interactive Trajectory Prediction. *International Conference on Intelligent Robots and Systems (IROS)*, 2022.
12. **Yaru Niu**. Adaptable and Scalable Multi-Agent Graph-Attention Communication. *Master's Thesis, Georgia Institute of Technology*, 2022.
13. Rohan Paleja*, **Yaru Niu***, Andrew Silva, Chace Ritchie, Sugju Choi, Matthew Gombolay. Learning Interpretable, High-Performing Policies for Autonomous Driving. *Robotics: Science and Systems (RSS)*, 2022.
14. **Yaru Niu***, Rohan Paleja*, Matthew Gombolay. Multi-Agent Graph-Attention Communication and Teaming. *International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, 2021 (**Oral**).
15. **Yaru Niu***, Rohan Paleja*, Matthew Gombolay. MAGIC: Multi-Agent Graph-Attention Communication. *Mair2 Workshop at International Conference on Computer Vision (ICCV)*, 2021 (**Best Paper Award**).
16. Zhijun Zhang*, **Yaru Niu***, Ziyi Yan, Shuyang Lin. Real-time Whole-body Imitation by Humanoid Robots and Task-oriented Teleoperation Using an Analytical Mapping Method and Quantitative Evaluation. *Applied Sciences (Special Issue Human-Friendly Robotics, Impact Factor: 2.217)*, 2018.
17. Zhijun Zhang, **Yaru Niu**, Shangen Wu, Shuyang Lin, Lingdong Kong. Analysis of Influencing Factors on Humanoid Robots' Emotion Expressions by Body Language. *International Symposium on Neural Networks (ISNN), Lecture Notes in Computer Science (LNCS)*, Springer, 2018.
18. Zhijun Zhang, Lingdong Kong, **Yaru Niu**. A Time-Varying-Constrained Motion Generation Scheme for Humanoid Robot Arms. *International Symposium on Neural Networks (ISNN), Lecture Notes in Computer Science (LNCS)*, Springer, 2018.

Preprints

(* indicates equal contributions)

1. **Yaru Niu**, Zhenlong Fang, Binghong Chen, Shuai Zhou, Revanth Senthilkumaran, Hao Zhang, Bingqing Chen, Chen Qiu, H. Eric Tseng, Jonathan Francis, Ding Zhao. Learning Versatile Humanoid Manipulation with Touch Dreaming. *In submission*, 2026.
2. Hongyi Chen, Tony Dong, Tiancheng Wu, Liquan Wang, Yash Jangir, **Yaru Niu**, Yufei Ye, Homanga Bharadhwaj, Zackory Erickson, Jeffrey Ichnowski. Dexterous Manipulation Policies from RGB Human Videos via 3D Hand-Object Trajectory Reconstruction. *In submission*, 2026.
3. Dongsu Lee*, **Yaru Niu***, Yongkang Liu, Ding Zhao. Vector-quantized Graph-attention Transformer for Multi-agent Communication. *In submission*, 2025.
4. Rohan Paleja*, Letian Chen* **Yaru Niu***, Andrew Silva, Zhaoxin Li, Songan Zhang, Chace Ritchie, Sugju Choi, Kimberlee Chestnut Chang, Hongtei Eric Tseng, Yan Wang, Subramanya Nagesh Rao, Matthew Gombolay. Learning Interpretable, High-Performing Policies for Continuous Control. Submitted to *Journal of Machine Learning Research (JMLR)*, 2023.
5. Mengdi Xu*, Peide Huang*, Wenhao Yu, Shiqi Liu, Xilun Zhang, **Yaru Niu**, Tingnan Zhang, Fei Xia, Jie Tan, Ding Zhao. Creative Robot Tool Use with Large Language Models. *arXiv Preprint*, 2023.
6. Lingdong Kong, Shaoyuan Xie, Hanjiang Hu, **Yaru Niu**, et al. The RoboDrive Challenge: Drive Anytime Anywhere in Any Condition. *arXiv Preprint*, 2024.
7. Lingdong Kong, **Yaru Niu**, Shaoyuan Xie, Hanjiang Hu, et al. The RoboDepth Challenge: Methods and Advancements Towards Robust Depth Estimation. *arXiv Preprint*, 2023.
8. Zhijun Zhang (PI), Lingdong Kong, **Yaru Niu**, Ziyang Liang. Modification of Gesture-Determined-Dynamic Function with Consideration of Margins for Motion Planning of Humanoid Robots. *arXiv Preprint*, 2020.

Patents

1. Zhijun Zhang, **Yaru Niu**. A Mapping Method of Human Postures Applied to Motion Imitation by Humanoid Robots (Translated from Chinese). *Published Authorization Number: CN107953331B*.
2. Zhijun Zhang, **Yaru Niu**. A Similarity Evaluation Method of Imitation by Humanoid Robots (Translated from Chinese). *Published Authorization Number: CN107818318B*.
3. Zhijun Zhang, **Yaru Niu**, Hao Wang. A Mapping Method of Human Body's Rotation and Displacement Applied to Humanoid Robots (Translated from Chinese). *Published Authorization Number: CN108858188B*.
4. Zhijun Zhang, **Yaru Niu**, Hao Wang. An Evaluation Metric of Humanoid Robot and Human Posture Similarity (Translated from Chinese). *Published Application Number: CN109064486A*.

Research Experience

Bosch Center for Artificial Intelligence <i>Machine Learning Research Intern, Robot Learning Lab</i>	May 2025 – Aug. 2025 <i>Mentors: Bingqing Chen, Jonathan Francis, Chen Qiu</i>
Bosch Center for Artificial Intelligence <i>Machine Learning Research Intern, Robot Learning Lab</i>	May 2024 – Aug. 2024 <i>Mentors: Bingqing Chen, Jonathan Francis, Zhenzhen Li</i>
Baidu Research <i>Research Intern, Robotics and Autonomous Driving Lab</i>	Jan. 2022 – July 2022 <i>Mentor: Liangjun Zhang</i>
University of California, Berkeley <i>Research Intern, Mechanical Systems Control Lab</i>	July 2021 – Feb. 2022 <i>Advisor: Masayoshi Tomizuka</i>
Georgia Institute of Technology <i>Graduate Research Assistant, Cognitive Optimization and Relational Robotics Lab</i>	Jan. 2020 – May 2022 <i>Advisor: Matthew Gombolay</i>
South China University of Technology <i>Undergraduate Researcher, Bionic Intelligent Robot Lab</i>	Aug. 2016 – June 2019 <i>Advisor: Zhijun Zhang</i>
University of California, Irvine <i>Undergraduate Researcher, Advanced Integrated Cyber-Physical Systems Lab</i>	Summer 2018 <i>Advisor: Al Faruque</i>

Honors & Awards

Georgia Tech ECE Fellowship	Feb. 2022
Best Paper Award, ICCV 2021 Mair2 Workshop (top 1)	Oct. 2021
AAMAS 2021 Scholarship	Feb. 2021
National Motivational Scholarship (twice) , awarded by Ministry of Edu. of China (rank 2/51)	Nov. 2016, 2018
The Jetta Scholarship , awarded by Jetta Company Limited (rank 3/51)	Dec. 2017
SCUT Scholarship , awarded by SCUT (rank 3/51)	Nov. 2017
2nd Prize in China Undergrad. Math. Contest in Modeling (CUMCM) (top 5% in SCUT)	Oct. 2017
1st Prize of Guangdong Province in CUMCM (top 5% in SCUT)	Oct. 2017

Teaching Experience

24784 Trustworthy Artificial Intelligence <i>Graduate Teaching Assistant, Carnegie Mellon University</i>	Jan. 2024 – May 2024
CS 4731/7632 Game Artificial Intelligence <i>Graduate Teaching Assistant, Georgia Institute of Technology</i>	June 2020 – Dec. 2020
CS 4641 Machine Learning <i>Graduate Teaching Assistant, Georgia Institute of Technology</i>	Jan. 2021 – May 2021

Mentorship

Revanth Senthilkumaran (MS at CMU)	Aug. 2025 – Present
Binghong Chen (MS at CMU)	Aug. 2025 – Present
Shuai Zhou (Visiting Researcher at CMU → MS at CMU)	Aug. 2025 – Present
Zhenlong Fang (Visiting Researcher at CMU → PhD at Berkeley)	Aug. 2025 – Present
Ryan Wang (Undergrad. at CMU)	Aug. 2025 – Present
Chenhao Li (MS at CMU)	Jan. 2025 – June 2025
Mingyang Yu (MS at CMU → Sharpa)	Aug. 2024 – Mar. 2025
Dongsu Lee (MS at Soongsil U. → PhD at UT Austin)	Aug. 2024 – Mar. 2025
Chuye Hong (Undergrad. at THU → PhD at Berkeley)	Summer 2024
Yuming Feng (Undergrad. at PKU → MS at Stanford)	Summer 2024
Yunzhe Zhang (Undergrad. at THU → PhD at NYU Courant)	Feb. 2024 – Feb. 2025
Ziang Cao (Undergrad. at U. of Pittsburgh → MS at Stanford)	Summer 2023

Academic Service

Reviewer: RA-Letters, CoRL, ICRA, IROS, WAFR, MRS, ICLR, NeurIPS, ICML, AISTATS

Program Committee: NeurIPS 2023-2024 Workshop on Foundation Models and Decision Making (FMDM), CoRL 2023 Workshop on Language and Robot Learning (LangRob), Machine Learning for Autonomous Driving (ML4AD) Symposium 2023, NeurIPS 2022 Workshop on ML4AD

Organizer: ICRA 2024 RoboDrive Challenge, ICRA 2023 RoboDepth Challenge